

JIFENG WU

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EDUCATION

CORNELL UNIVERSITY

Ithaca, NY

PhD Student

AUGUST 2025 - EXPECTED JUNE 2031

UNIVERSITY OF BRITISH COLUMBIA

Vancouver, Canada

MSc in Computer Science

NOVEMBER 2024

- Advisor: Caroline Lemieux
- GPA 4.0, Coursework: Software Engineering, Automated Testing, Type Systems, Deep Learning

WUHAN UNIVERSITY

Wuhan, China

BEng in Software Engineering, Outstanding Graduate

JUNE 2022

- Advisor: Qingan Li
- GPA 3.93, Coursework: Operating Systems, Database Systems, Software Engineering, Embedded Software Design, Computer Systems: A Programmer's Perspective, Compilation Technology and Application

PUBLICATIONS

QuAC: Quick Attribute-Centric Type Inference for Python

Jifeng Wu, Caroline Lemieux

Proceedings of the ACM on Programming Languages, Issue OOPSLA, 2024

Effective Stack Wear Leveling for NVM

Jifeng Wu, Wei Li, Libing Wu, Mengting Yuan, Chun Jason Xue, Jingling Xue, Qingan Li

IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2023

RESEARCH EXPERIENCE

UNIVERSITY OF BRITISH COLUMBIA

Vancouver, Canada

QuAC: Quick Attribute-Centric Type Inference for Python

JANUARY 2023 - AUGUST 2024

- Implemented QuAC, a novel Python type inference tool using attribute sets and information retrieval.
- Demonstrated high coverage and accuracy, outperforming non-LLM baselines, especially in predicting container type parameters and non-built-in types, while significantly reducing run times.
- Compared to LLM-based methods, QuAC is nearly two orders of magnitude faster while achieving greater consistency in its container type parameter predictions.
- GitHub: github.com/jifengwu2k/quac

WUHAN UNIVERSITY

Wuhan, China

Effective Stack Wear Leveling for NVM

AUGUST 2021 - AUGUST 2022

- Proposed Loop2Recursion, a stack wear leveling technique implemented as an LLVM pass for increasing the lifespan of NVM with limited write durability by converting wear-heavy loops into recursive functions.
- Outperforms state-of-the-art methods by significantly improving stack wear leveling and reducing performance overhead.
- GitHub: github.com/jifengwu2k/loop2recursion

WUHAN UNIVERSITY

Wuhan, China

Community Detection Using Social Networks and Trajectories

SEPTEMBER 2019 - AUGUST 2021

- Explored using trajectories to identify spatiotemporally cohesive user groups in large social networks.
- Developed a linear-time spatiotemporal trajectory similarity algorithm and community detection algorithm.
- Comprehensive evaluations on two datasets demonstrated the effectiveness and efficiency of these algorithms.

TEACHING EXPERIENCE

CORNELL UNIVERSITY

Ithaca, NY

Teaching Assistant, ECE 2400: Computer Systems Programming

JANUARY 2020 - APRIL 2020

- Supported students in computer systems programming and C/C++ development through technical guidance, debugging help, and course support.
- Developed assignment infrastructure, including repository management tools, assignment releases, grading workflows, final grading scripts, and GitHub Actions self-checking.
- Delivered section presentations on Unix tooling, C++ environment setup, and C++ profiling.

UNIVERSITY OF BRITISH COLUMBIA

Vancouver, Canada

Graduate Teaching Assistant, CPSC 410: Advanced Software Engineering

SEPTEMBER 2022 - DECEMBER 2022

- Supported students on static and dynamic program analysis course projects.
- Provided technical feedback on project design, implementation, and analysis strategies.
- Helped students understand advanced programming language and software engineering concepts through individualized guidance.

OPEN-SOURCE CONTRIBUTIONS

AVL ORDER STATISTIC SET

C++11

- A C++11 header-only self-balancing AVL set with $O(\log n)$ order-statistics/range queries.
- API-compatible with `std::set`, portable replacement for GNU PBDS.
- GitHub: github.com/jifengwu2k/avl-order-statistic-set

CHATREPL

Python 2+

- An agent for OpenAI-compatible Chat Completions APIs featuring 4 tools: read, write, edit, shell.
- Supports streaming responses, multiline input via `$EDITOR`, and file integration.
- PyPI: pypi.org/project/chatrepl

COWLIST

Python 2+

- An immutable, hashable, copy-on-write list-like data structure implementing the Sequence protocol.
- Supports $O(1)$ indexing/slicing/appending-at-end and structural sharing.
- PyPI: pypi.org/project/cowlist

COWTREEDICT

Python 2+

- An immutable, hashable, copy-on-write sorted dictionary backed by a balanced tree.
- Supports $O(\log n)$ functional updates, structural sharing, and order-statistic queries such as rank and selection.
- PyPI: pypi.org/project/cowtreedict

PYSSA

Python 3.6+

- An executable, source-oriented, stable intermediate representation that represents Python programs in an SSA-like form, enabling custom execution, static analysis, and program transformation.
- GitHub: github.com/jifengwu2k/pyssa

SKILLS

- Languages: Python, C/C++, Bash.
- Frameworks: PyTorch, NetworkX, PyQt/PySide, LLVM, MLIR, Intel Pin, Git, Docker.
- Domains: Program Analysis, Formal Methods, Compilers, Automated Reasoning.

CERTIFICATIONS

Google AI Professional Certificate

MAY 2026

- Completed seven-course program covering AI literacy, responsible AI use, prompting, AI-assisted research, writing, data analysis, content creation, and app building.

AWARDS

- 3x Outstanding Student Scholarship (Top 5%)
- 2x Advanced Individual in Social Work
- China Software Cup (2020)
- Service Outsourcing Innovation and Entrepreneurship Competition for Chinese College Students (2021)